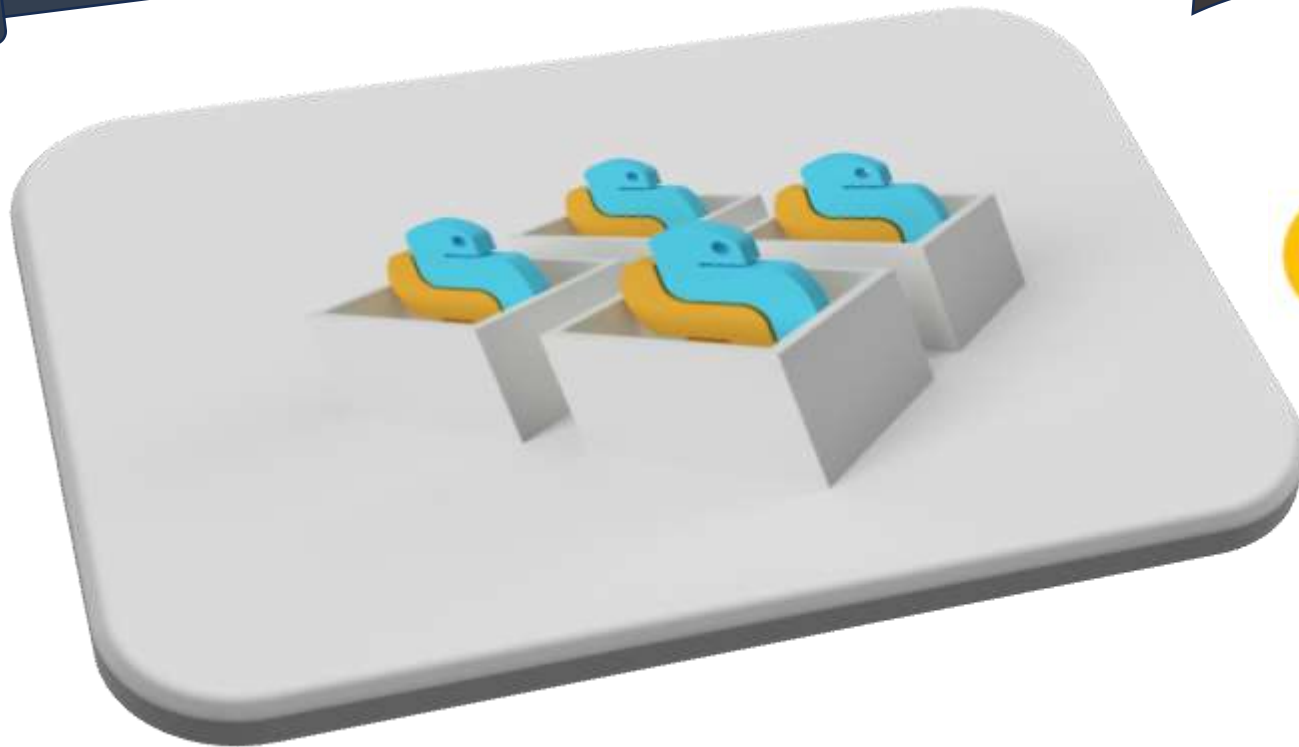




Your First Step into the Python World



Setting Up the Environment (Google Colab, Anaconda) & Writing Your First Python Program

colab

 jupyter



➤ Today's Learning Journey

□ Map for Today:

1. 🌐 Google Colab – Your online coding notebook
2. 🐍 Anaconda – Your offline Python powerpack
3. 📓 Jupyter Notebook – Your lab book
4. 💻 VS Code – The modern code editor
5. 📖 Writing & Running Your First Python Program



🎯 *"By the end of this class, you'll be a Python Pilot!"*



➤ Python Environments – The Big Picture

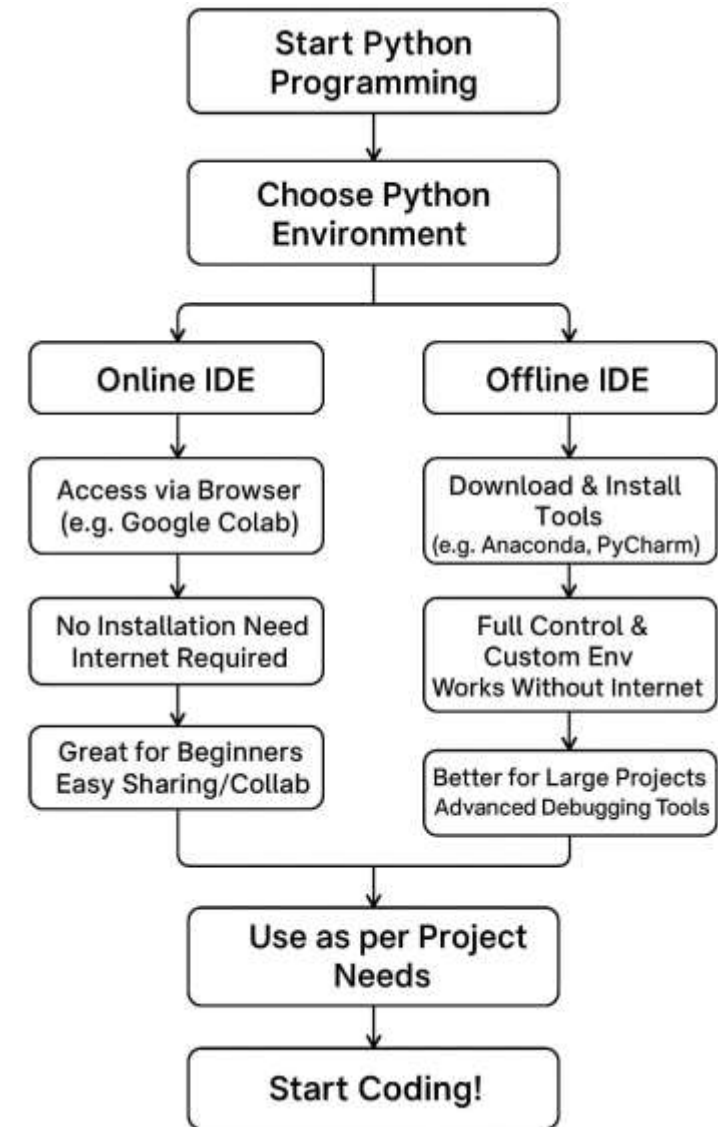
❑ Two Roads to Start Your Python Journey:

🔗 **Online** – Google Colab (no install, access anywhere)

🔗 **Offline** – Anaconda (powerful tools in one pack)

❑ **Online = Cloud Coding**

🔌 **Offline = Local Setup (your machine)**



➤ Google Colab – Your Free Online Lab

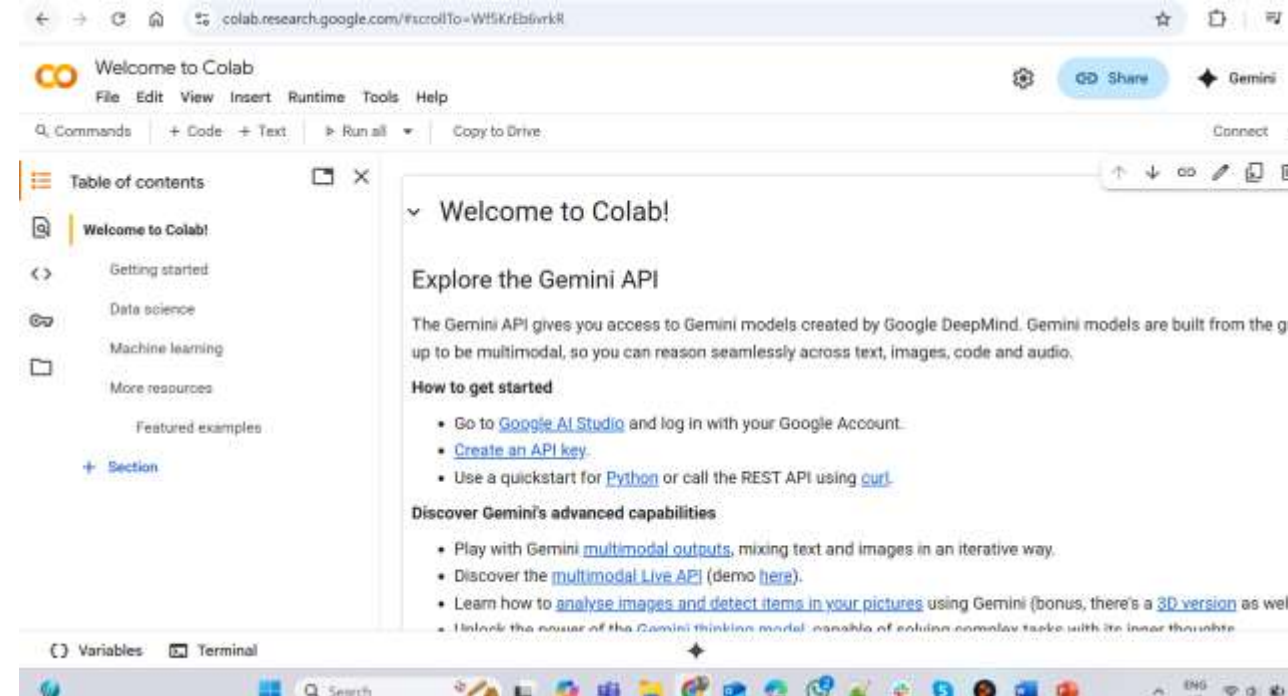
☁ Think of Colab as Google Docs for Python!

📁 Benefits:

- Runs in your browser 🌐
- No installation ☐
- Auto-saves on Google Drive ☐
- Cell-based execution → Easy testing

☐ Try it now:

🔗 <https://colab.research.google.com>



➤ Let's Launch Colab 🚀

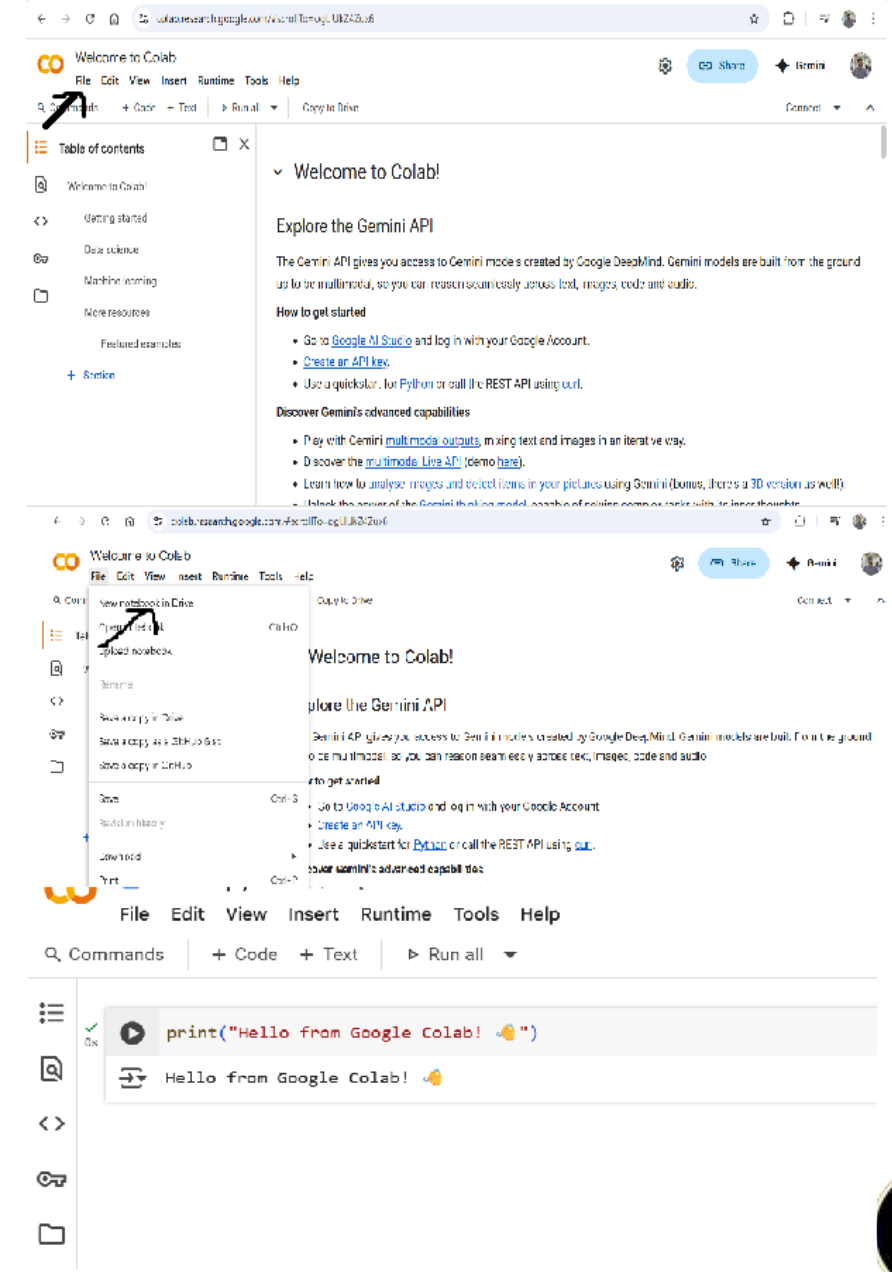
Step-by-Step to Run Your First Cell:

1. Open Colab
2. Click “New Notebook”
3. Type this:

```
print("Hello from Google Colab! 🖐️")
```

4. Press Shift + Enter

🎯 *You just executed your first Python code on the cloud!*



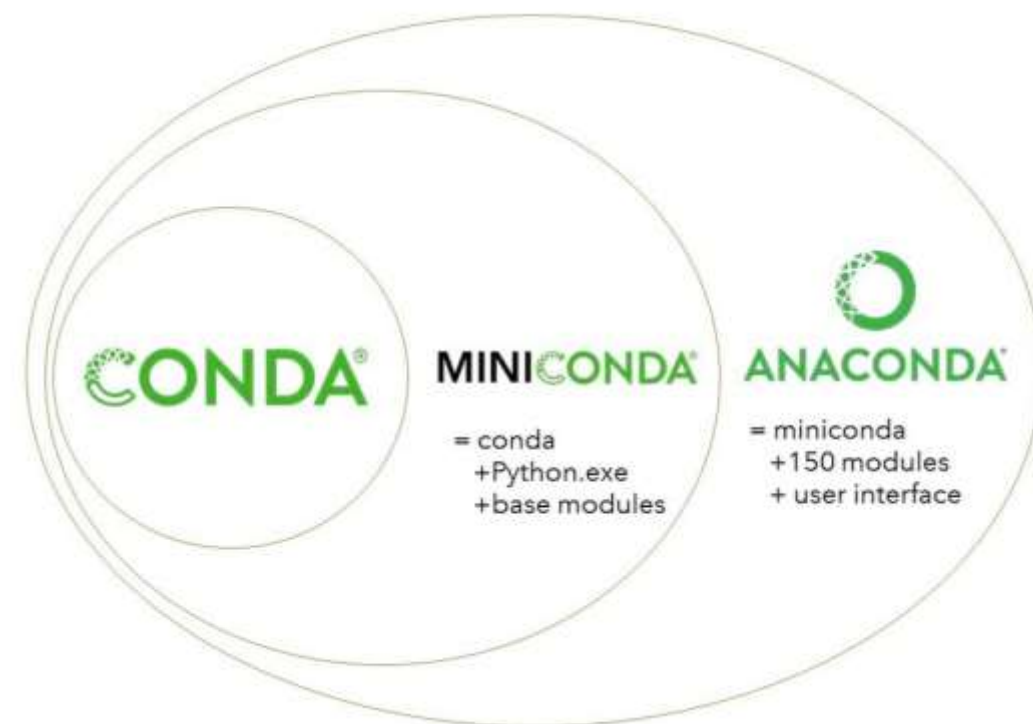
➤ What is Anaconda? – Python's Power Suite

□ **Anaconda = Swiss Army Knife for Python Devs**

It includes:

- Python itself 🐍
- Jupyter Notebook 📓
- Conda (library manager) □
- Spyder (IDE) 🖥️

📦 All bundled neatly for data science, ML, and automation

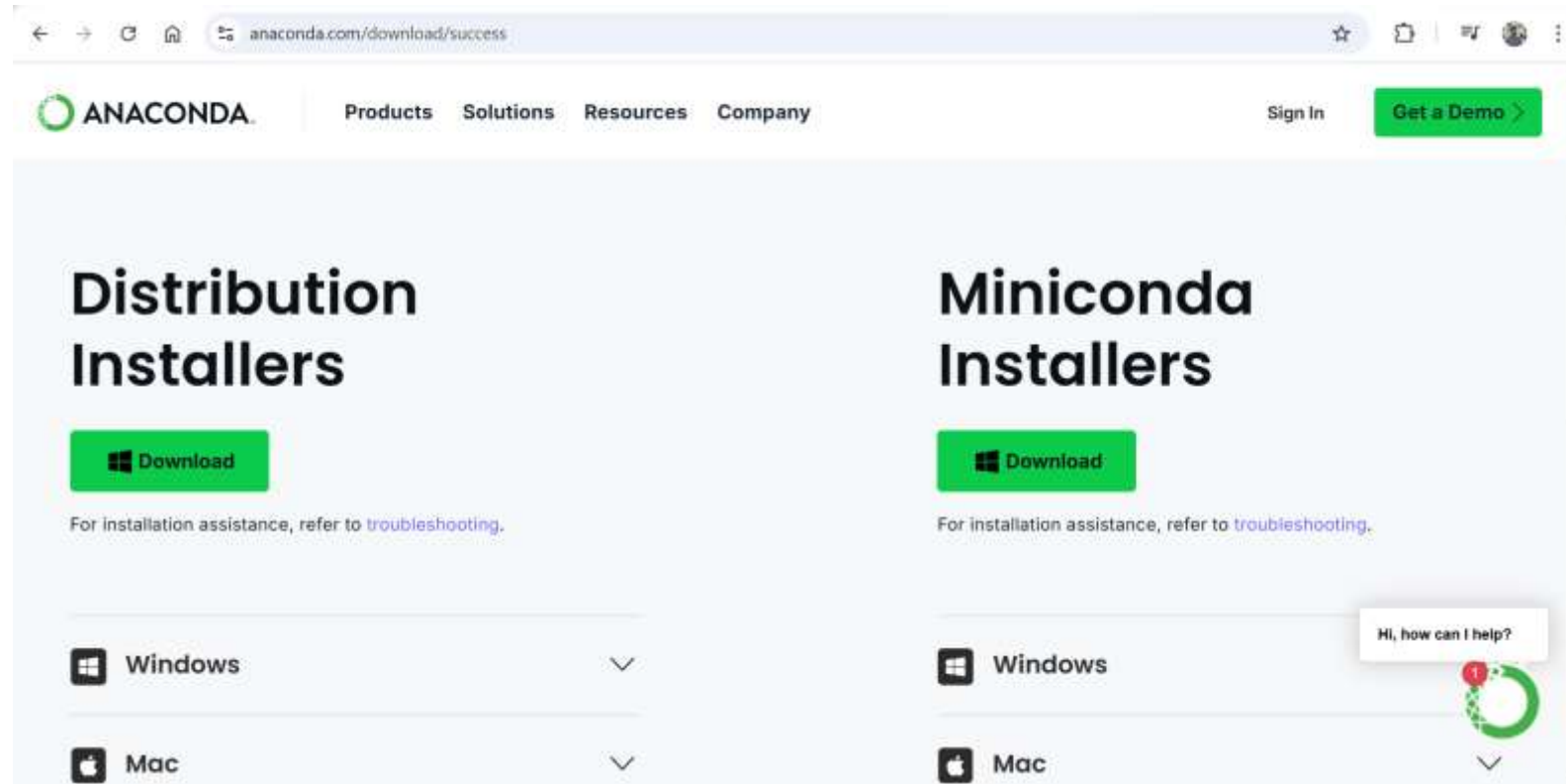


➤ Installing Anaconda

☐ 🔧 Let's Gear Up:

1. Go to anaconda.com
2. Download for your OS
3. Install with default options
4. Search and open:
 1. 📁 Anaconda Navigator
 2. 📄 Jupyter Notebook

✓ *You're now Python-ready offline!*



➤ Meet Jupyter – Your Digital Lab Notebook

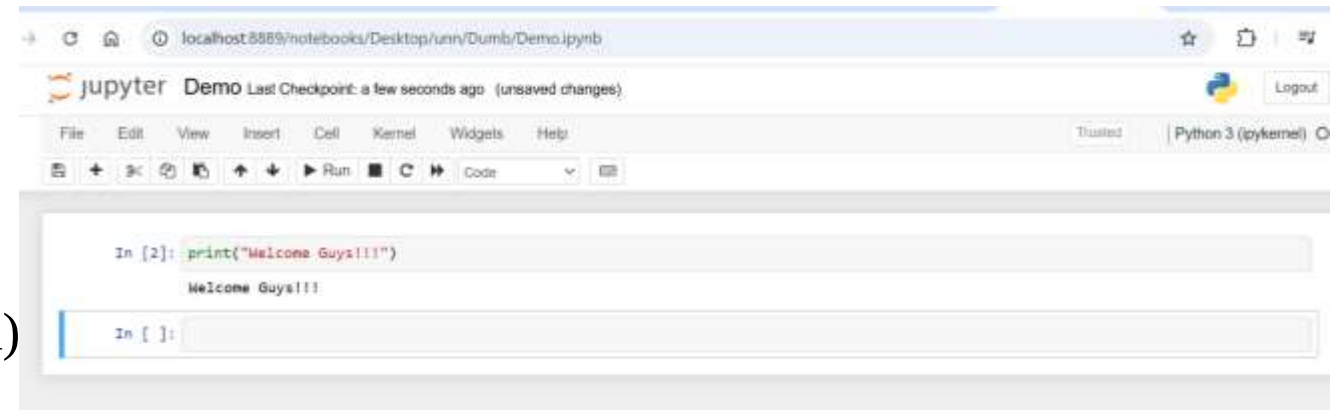
📖 Jupyter = Notebook where code & notes live together

✦ Features:

- Cell-by-cell execution
- Markdown support (for text, images, math)
- Save .ipynb files
- Run directly in browser (localhost)

💡 Real Life Use:

- Used by Data Scientists, Researchers & AI Experts



➤ VS Code

□ VS Code = Python with Superpowers

Why use it?

- ☁ Beautiful interface
- □ Smart code suggestions
- ↻ Git & Terminal support
- 🍴 Extensions for Python, Django, Flask

📁 You'll need:

- Python installed
- VS Code + Python extension



➤ .py vs .ipynb – What's the Difference?

Feature	.py (Python File)	.ipynb (IPython Notebook File)
Full Form	Python File	Interactive Python Notebook
File Extension	.py	.ipynb
Used In	Text editors, IDEs (VS Code, PyCharm, etc.)	Jupyter Notebook, Google Colab
File Format	Plain text	JSON (structured as cells)
Structure	Sequential code	Cell-based (Code, Markdown, Output)
Execution Style	Entire file runs at once	Code runs cell by cell
Supports Markdown/Text	✗ No	✓ Yes
Supports Visualization	✓ Yes (via libraries like Matplotlib, Seaborn)	✓ Yes (inline and interactive)
Best For	Script writing, production code	Data science, teaching, tutorials
Interactivity	Low	High (interactive and exploratory)
Ease of Version Control	✓ Easy (diffs are clean in Git)	✗ Difficult (JSON format can be messy in diffs)
Popular Tools	Python interpreter, command line, IDEs	Jupyter Notebook, Google Colab
Beginner Friendly	Moderate	✓ Very (step-by-step execution, visual feedback)



➤ Your First Program 🎯

Let's write your **first Python code** ever:

```
print("Hello, Python World! 🎯")
```

💬 Explanation:

- `print()` → Function that displays text
- `"..."` → Anything inside quotes is a **string**

🔗 Result:

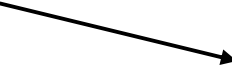
```
Hello, Python World! 🎯
```

✓ Try it in Colab, Jupyter, or VS Code



➤ Let's Interact – Input + Output

```
name = input("What's your name? ")  
print("Welcome,", name)
```



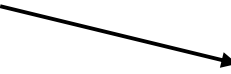
```
C:\Users\RITESH\Desktop\unn\JAVA>python hello.py  
What's your name? 22  
Welcome, 22
```

✦ This program:
Accepts input from user

Greets the user dynamically!

🎯 Fun Practice:

```
age = int(input("Enter your age: "))  
print("Next year, you'll be", age + 1)
```



```
C:\Users\RITESH\Desktop\unn\JAVA>python hello.py  
Enter your age: 22  
Next year, you'll be 23
```



➤ Recap & Takeaway

□ Today You Learned:

- Google Colab – Cloud coding
- Anaconda & Jupyter – Offline setup
- .py vs .ipynb
- First Python program with print()
- Taking input using input()

🏆 *You are now ready to build your Python foundation!*



